AI ACADEMY · LEVEL 6 · GRADE 11 · WEEK 17 · STUDENT LESSON

# External side effects and recovery

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

## Before you start

Recall Timeouts retries and cancellation. Explain why a recorded result needs its input conditions and evidence source before it can support a claim.

Plan time to read, investigate and explain. Keep predictions separate from observations.

STUDENT LESSON · 1 OF 18

## Your mission

> Start with the target: Explain the mechanism and terminology of external side effects and recovery. Try the recall before opening notes.

### Your learning target

I can explain the **mechanism** and terminology of external side effects and recovery.

I can apply the stated procedure to a new case with a traceable result.

I can identify an assumption or limitation and design a revealing follow-up.

### Remember before reading

Close your notes first. A rough first answer helps us choose the right starting point; it is not a score for your ability.

Without opening the old lesson, explain one key idea from Timeouts retries and cancellation. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Planning and action selection. Give an example and a mistake that the idea helps you avoid.

Without opening the old lesson, explain one key idea from Failure budgets and service objectives. Give an example and a mistake that the idea helps you avoid.

After trying, check the relevant lesson or ask your teacher. Change the part of your explanation that the **evidence** shows was wrong.

This week you will practise:

Explain the **mechanism** and terminology of external side effects and recovery

Apply the stated procedure to a new case with a traceable result

Identify an assumption or limitation and design a revealing follow-up

### Retrieve what you already know

Recall Timeouts retries and cancellation. Explain why a recorded result needs its **input** conditions and **evidence** source before it can support a claim.

> **My starting explanation**

### This week's end goal

Explain external side effects and recovery using the deciding step in this fictional service.

## Notice the evidence

> Investigate the supplied External side effects and recovery case. Separate observations, given facts and assumptions before choosing an explanation.

External effects can succeed while their **acknowledgement** is lost. Recovery must inspect committed state before repeating the operation.

Begin by writing the question the procedure is intended to answer. Identify the information already supplied, the information it will produce and the information still missing.

These are constructed classroom examples; they do not require live accounts, private records or an external service. Do not treat a plausible explanation as a measured outcome.

**List the given facts and one fact that is not supplied.**

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## See how it works

> Follow the mechanism for External side effects and recovery. Connect each labelled input, deciding step and result.

The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action.

### External side effects and recovery — evidence trace

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure. Preserve compensation as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost acknowledgement does not undo the effect. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Before applying the procedure, write its units, denominator or decision boundary where relevant. A partner should be able to repeat each step using the same **input**s.

If a required **input** is absent or violates the stated contract, record that condition explicitly instead of quietly replacing it with a convenient value.

## Connect the picture and the explanation

Point to the **input**, the deciding step and the result in this week’s visual. Explain the same step in ordinary words. A picture helps when you can say what each part represents.

# Compare the cases

> Compare the supplied cases for External side effects and recovery. Identify the detail that changes the conclusion.

### Compare a familiar case with a changed case

A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

> **Keep the procedure fixed. Explain what changed in the inputs and result.**

### Change one thing in the pictured case

#### Change one thing

Change: Status reliably reports J6 absent.

#### Predicted result

**Prediction**: A policy-approved resend using the same deduplication key may be considered.

#### Why the result follows

Reason: Confirmed absence supplies different **evidence** from silence.

## Words that unlock the idea

> Use the vocabulary of External side effects and recovery to explain a concrete decision, rather than only memorising definitions.

| Word | Meaning |
| --- | --- |
| Acknowledgement | A response confirming operation status |
| Reconciliation | Checking authoritative state after uncertainty |
| Compensation | A separate action intended to offset a prior effect |

> **Use two terms to explain a step in this week's example.**

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STUDENT LESSON · 6 OF 18

# Follow a worked example

> Follow the worked trace for External side effects and recovery, predict a step, then compare with its explanation.

A booking is committed in storage, but response delivery times out. Caller asks to repeat.

Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

### Worked explanation · External side effects and recovery

Compare the intermediate **evidence** with the final statement. Explain why each number or decision follows from the stated **input**s.

The conclusion belongs to this scenario and procedure; changing the **input**s, evaluation population or authority can change what it establishes. Retain the raw record so a later revision can be compared honestly.

#### Learn from the worked case

Cover the result before checking it. Say what is given, choose the procedure, and follow one step at a time. Then uncover the result and explain your first mismatch. Ask why a step is allowed, not only what number it produces.

#### A pictorial worked case: External side effects and recovery

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

#### The facts we will use

### Supplied facts

Job key J6 identifies one badge. The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

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# External side effects and recovery

**Start with the supplied facts**

Follow each arrow. Ask which fact allows the next step.



## 1. Record uncertainty

Sending occurred, but printing completion is not yet known.



## 2. Query J6

Ask for the existing job's recorded state.



## 3. Recover from status

If complete, return that result without issuing another print job.

**CHANGE ONE THING**

Status reliably reports J6 absent.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

### Read the picture one step at a time

| STEP | WHAT HAPPENS AND WHY |
| --- | --- |
| 1. Record uncertainty | Sending occurred, but printing completion is not yet known. |
| 2. Query J6 | Ask for the existing job’s recorded state. |
| 3. Recover from status | If complete, return that result without issuing another print job. |

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

### Why this result follows

#### Why the result follows

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

### What this example does not establish

#### Keep the stated limit

Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

### Keep the stated limit

Read the labelled picture: Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

### Keep the stated limit

Change one thing: Status reliably reports J6 absent. Predict the result before checking the explanation. What remains the same?

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## Find and repair a mistake

> Find the unsupported step in the External side effects and recovery claim and explain a repair using the supplied evidence.

A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

### Check the claim against the facts

A useful check asks what would make the reasoning fail. Distinguish a failure to execute the declared procedure from a weakness in the procedure itself.

The first may require a correction; the second requires an explicitly revised method and new **evidence**. Preserve unsuccessful cases so an improved result cannot hide the original problem.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

### Locate the earliest unsupported step

A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

**State one repair and the checks you would rerun.**

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STUDENT LESSON · 8 OF 18

## Practise together

> Practise External side effects and recovery with support, then explain the deciding step in your own words.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

**Guided: restate the worked result, then name the assumption or limitation that prevents a broader conclusion.**

**Explain your prediction to a partner before comparing results.**

### Practise with less help: Trace the worked case

Use workbook W2 for the connected example “External side effects and recovery”. First fill the **input** and rule boxes. Try the middle steps before asking for a hint. After a hint, try the next step yourself.

| Plan | Do | Check |
| --- | --- | --- |
| What is given? Which rule or procedure applies? | Record each intermediate step. | Does the result follow? What remains unknown? |

Explain your trace to a partner. Your partner points to one step to check. Swap roles on the next case. Each person writes their own explanation.

### Check your reasoning

Use the worked case for External side effects and recovery. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Answer on your own before discussion. If you are unsure, say which step you need help with.

### Explain the picture

> Distinguish a missing printer **acknowledgement** from confirmed failure. What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

**My explanation using a pictured step and the given facts**

This is supported practice. Keep the separately named independent case for an attempt before feedback.

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STUDENT LESSON · 9 OF 18

## Run the investigation

> Investigate External side effects and recovery. Record the prediction, actual observation or labelled paper trace, explanation and next test separately.

### Predict, investigate and preserve the evidence

A booking is committed in storage, but response delivery times out. Caller asks to repeat. Show the intermediate steps and give the conclusion with its scope.

| Record | Your evidence |
|---|---|
| Given input and fixed rule | |
| Prediction before checking | |
| Actual result, or “not run” | |

| Record | Your evidence |
| --- | --- |
| First mismatch and one explanation | |
| Boundary or missing input and expected response | |

On paper, trace the supplied procedure. If code is provided, predict before running it in a local practice file. Never write an expected answer as though it were an observed run.

**Plan → try → check**

Before trying: what do I expect, and why? While trying: which step could first go wrong? Afterwards: what did I actually observe, and what should change? Keep “not run” if you only traced the procedure.

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## Build your understanding

> Move from recall to tracing, diagnosis and transfer in External side effects and recovery. Explain why each step is justified.

**Explain → apply → challenge**

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

A report omits this qualification: A compensating action is itself a side effect requiring authority; recovery does not silently permit deletion or refund.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**Explain which evidence supports your conclusion and which claim remains unproved.**

STUDENT LESSON · 11 OF 18

# Connect your project

> Apply this week's idea to Grounded Learning Service. Keep an annotated decision and the evidence supporting it.

Add a labelled external side effects and recovery **evidence** sheet to Grounded Learning Service. Show how you can explain the **mechanism** and terminology of external side effects and recovery.

Use the supplied fictional case and keep its **input**s, intermediate steps, calculation or **prediction**, and limitation. Connect this record to last week's record; a matching answer without a trace is insufficient.

| Project evidence | Record |
| --- | --- |
| Component and version | |
| Inputs permitted at prediction time | |
| Trace and comparison with earlier work | |
| Failure, limit and human review | |

## Evidence for your project

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

## Show what you know

> Use a genuinely new case for independent evidence. Previously discussed examples remain practice; use the separately named independent case before feedback.

Complete the supported workbook tasks before attempting W4 independently. Record any hints or help used. After feedback, use the teacher's fresh retry rather than copying the earlier answer.

For a project, select a relevant new case, predict the outcome from the declared rule and preserve the result alongside any revision.

**Exit: propose one revealing change to the worked scenario. State the expected outcome and the evidence that would make you revise your explanation.**

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

**My limited claim and a useful next check**

## Study a complete case

> Revisit the complete External side effects and recovery case for supported practice. Repeated examples do not become fresh assessments.

### Complete evidence record

**External side effects and recovery — evidence trace**

| INPUT AND SCOPE | CHECK OR CALCULATION | CONCLUSION AND LIMIT |
| --- | --- | --- |
| A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID. Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure. Preserve compensation as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost acknowledgement does not undo the effect. |

The figure connects this chapter's worked input, declared procedure and result. Read the complete worked reasoning for qualifiers and intermediate steps; the abbreviated labels do not replace that reasoning.

Follow the **input** through the stated procedure to its result. Cover the result first, predict it, then explain every intermediate step. The text/table accompanies the figure so the example can be read without colour.

**My complete trace of the worked case**

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## Try a new case independently

> Attempt the reserved new case before feedback. Record support; the earlier examples below remain additional practice.

### Independent case: Apply a new case independently

> A payment-like classroom token transfer has unknown status after network loss. Should the mock repeat immediately?

**My independent explanation, trace and limits**

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

### Attempt this without hints

> A payment-like classroom token transfer has unknown status after network loss. Should the

mock repeat immediately?

**My independent procedure, result and limitation**

Record support used: none, hint or teacher explanation. After feedback, a different case is needed to demonstrate independence.

### Use feedback to improve

Attempt the independent case before hints. Record whether you worked alone, used a hint or followed a model. After feedback, fix the first unsupported step and explain why it changed.

A later check uses a changed case, not a copied answer.

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## Your lesson route

> Use this map to navigate External side effects and recovery, connect the goals to tasks and keep one evidence record for each attempt.

Start with this question: Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

| By the end, I can | How I will show it |
| --- | --- |
| Explain the mechanism and terminology of external side effects and recovery | Explain the deciding step, show a trace or test, and state one limit. |
| Apply the stated procedure to a new case with a traceable result | Explain the deciding step, show a trace or test, and state one limit. |

| By the end, I can | How I will show it |
| --- | --- |
| Identify an assumption or limitation and design a revealing follow-up | Explain the deciding step, show a trace or test, and state one limit. |

Remember → watch and explain → try with help → investigate → try independently → keep project **evidence**. Your teacher may spread this lesson across several classes.

Keep your first **prediction**. Compare it with what the **evidence** shows and explain any change.

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## Finish and return

> Explain External side effects and recovery without notes, identify a remaining question and plan a later check with a changed case.

Close your notes. Explain this goal in your own words: Explain the **mechanism** and terminology of external side effects and recovery

**My deciding step, evidence and one remaining question**

If you are unsure, return to the worked picture and explain one arrow. If you are ready, change an **input** or assumption and justify your **prediction**.

Save your workbook **evidence**. At the next review, try a different case before opening your old answer.

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STUDENT LESSON · 17 OF 18

# Supported practice, repair and portfolio

> Attempt supported questions before using hints or feedback. Review a missing building block, then keep the completed work.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Recover an uncertain print job. A badge printer loses its reply after receiving a print request.

### Practice 1: explain the deciding evidence

Which labelled step matches this detail? If complete, return that result without issuing another print job.

1. Record uncertainty

2. Recover from status

3. Query J6

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: Recover from status If complete, return that result without issuing another print job. A lost response does not show whether an external effect occurred.

Recovery distinguishes unknown outcome from a confirmed failed action.

**Practice 2: explain the deciding evidence**

### Changed case · predict first

Changed-case practice: Status reliably reports J6 absent. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. A policy-approved resend using the same deduplication key may be considered.

3. We can decide the result without examining the changed **input** or the procedure.

**My selected conclusion, deciding fact and explanation**

### Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

### Feedback · compare after trying

After your attempt, compare with the supported explanation: A policy-approved resend using the same deduplication key may be considered. Confirmed absence supplies different **evidence** from silence.

**Repair a missing building block**

Review Timeouts retries and cancellation (ai-6/16). Goal: Explain the **mechanism** and terminology of timeouts retries and cancellation.

Short practice: Calculate waiting time for a one-second timeout followed by a 0.4-second success. What changes if cancellation arrives after the first attempt but before the retry?.

Return to this week and explain what you changed.

Review Planning and action selection (ai-6/14). Goal: Explain the **mechanism** and terminology of planning and action selection. Short practice: Choose a permitted plan when no approved tower-height card exists.

What new answer becomes supported after a card records 24 metres, and why could the bot not invent it earlier?. Return to this week and explain what you changed.

Review Failure budgets and service objectives (ai-6/11). Goal: Explain the **mechanism** and terminology of failure budgets and service objectives. Short practice: Calculate the maximum allowed failures among 100 requests.

Does 97 successes and 3 failures meet the target, and what changes after one failure becomes a success?. Return to this week and explain what you changed.

### Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

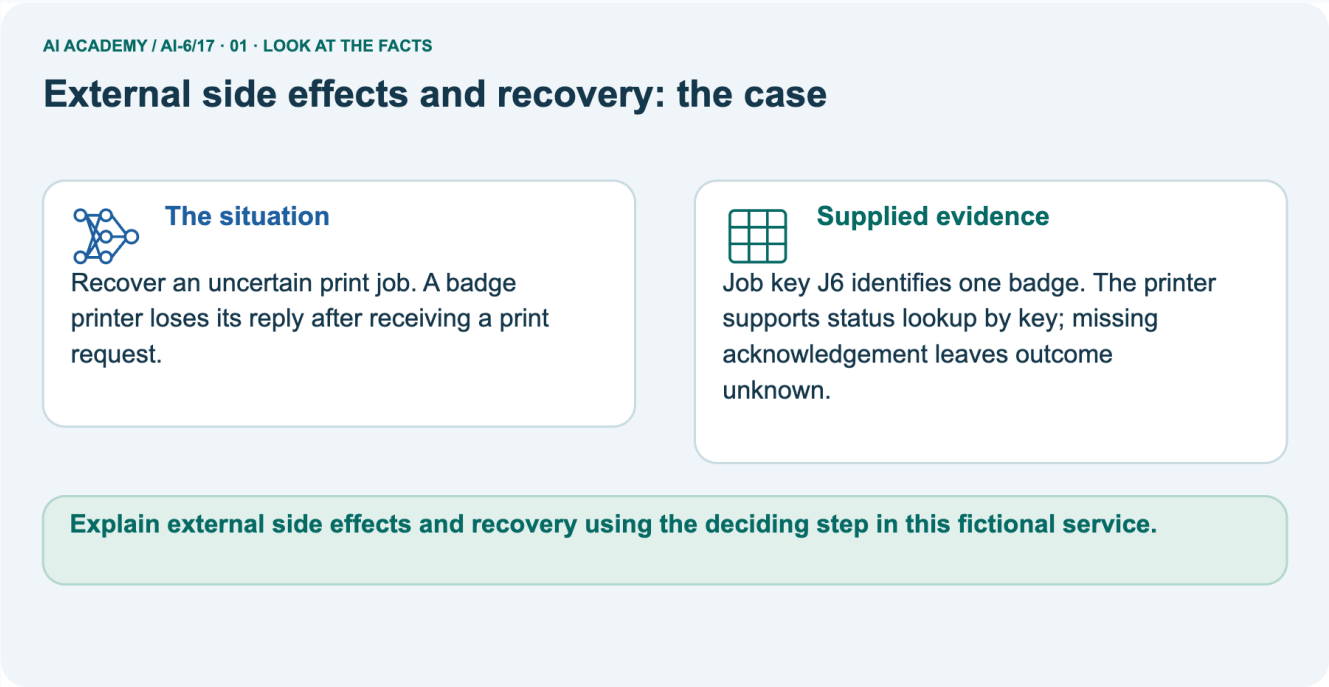
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## Visual concept journey

> Explain each External side effects and recovery graphic using its labels, deciding evidence and relevant limitation.

### 01. External side effects and recovery: the case



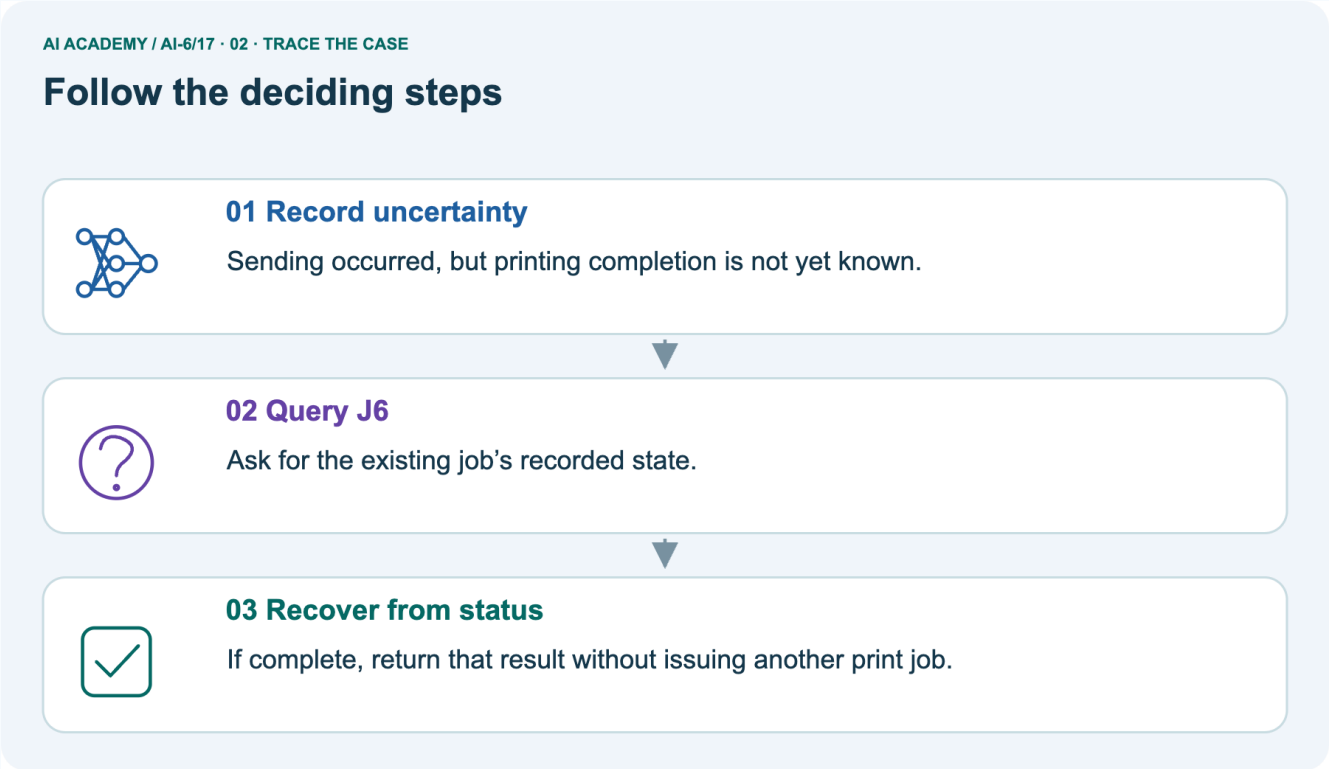

Follow the labelled picture with the matching explanation. Open the vector diagram

Recover an uncertain print job. A badge printer loses its reply after receiving a print request. Supplied facts: Job key J6 identifies one badge.

The printer supports status lookup by key; missing **acknowledgement** leaves outcome unknown.

> Point to the given fact that matters most. Explain why it matters.

### 02. Follow the deciding steps

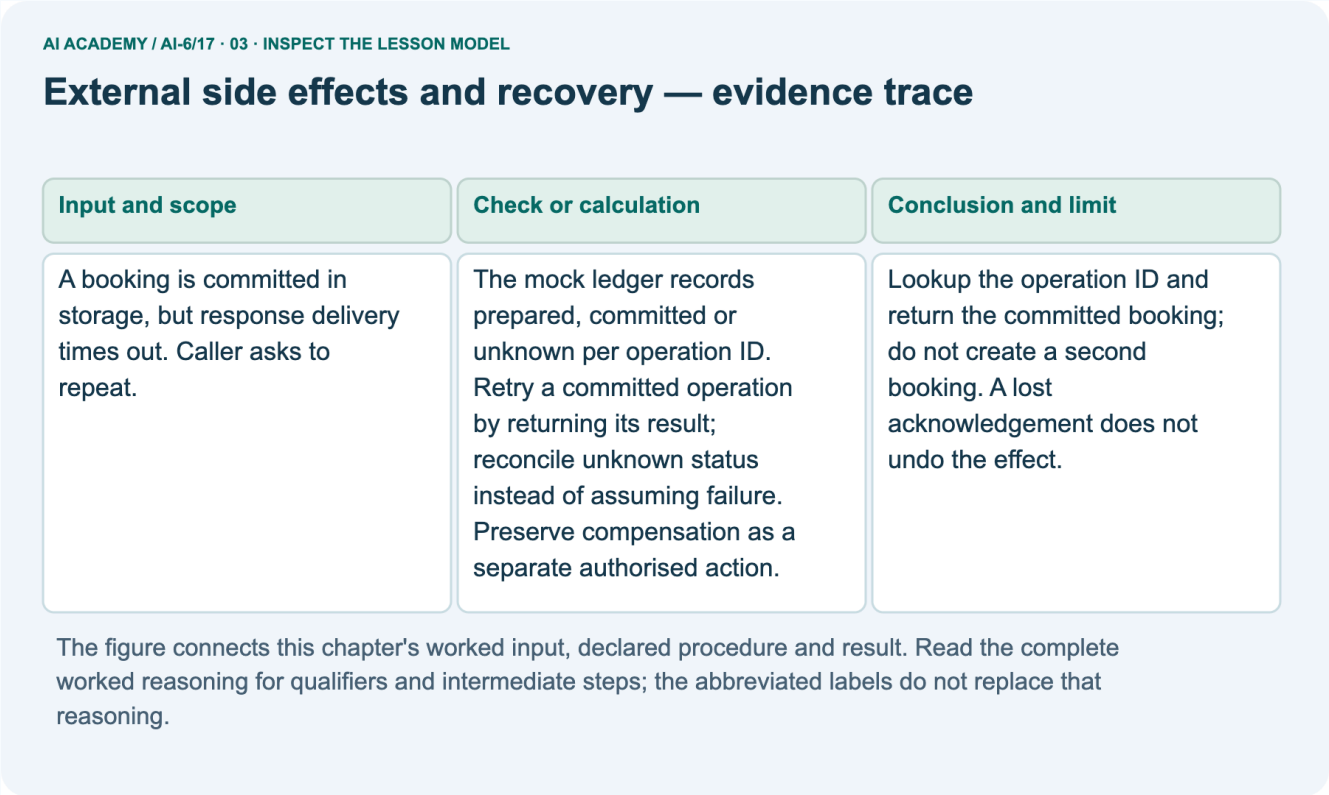



Follow the labelled picture with the matching explanation. Open the vector diagram

Record uncertainty: Sending occurred, but printing completion is not yet known. → Query J6: Ask for the existing job's recorded state. → Recover from status: If complete, return that result without issuing another print job.

> Point to each numbered stage and explain which recorded **input** it uses.

## 03. External side effects and recovery — evidence trace




Follow the labelled picture with the matching explanation. Open the vector diagram

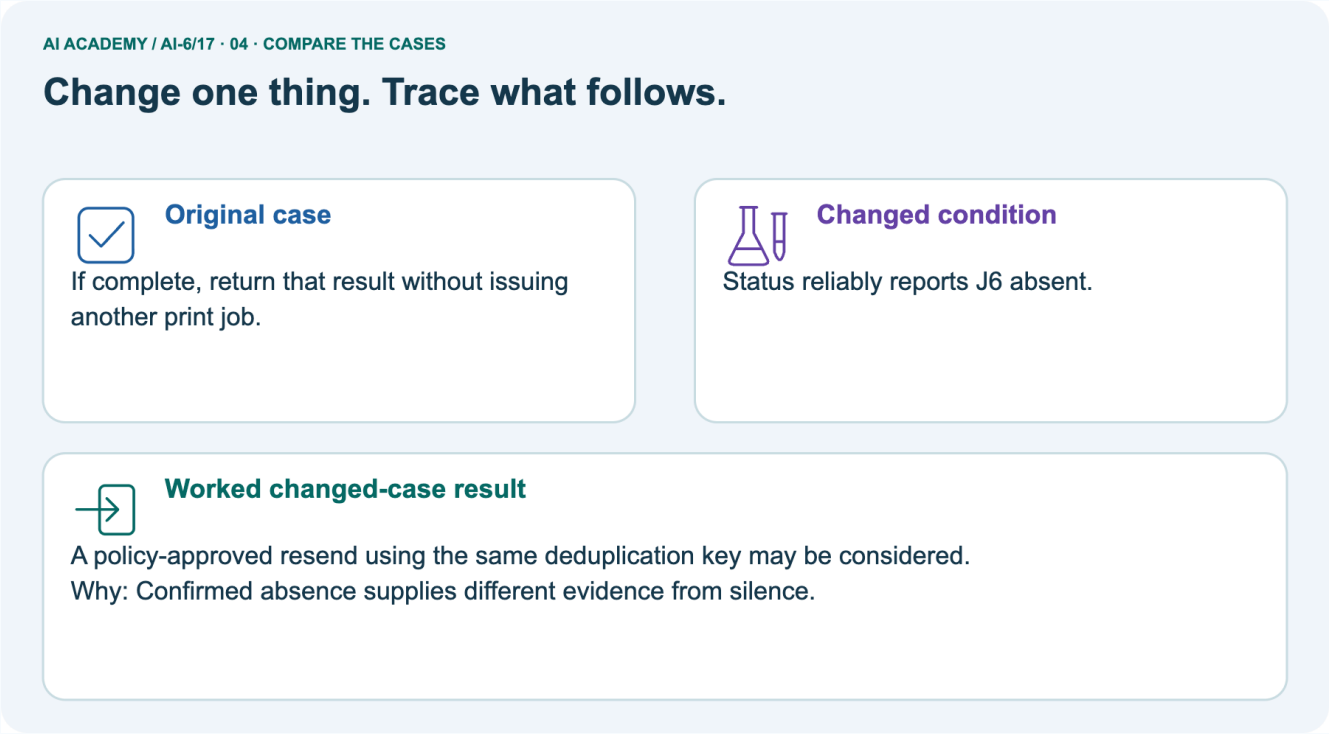
External side effects and recovery — **evidence** trace. A booking is committed in storage, but response delivery times out. Caller asks to repeat. | The mock ledger records prepared, committed or unknown per operation ID.

Retry a committed operation by returning its result; reconcile unknown status instead of assuming failure.

Preserve **compensation** as a separate authorised action. | Lookup the operation ID and return the committed booking; do not create a second booking. A lost **acknowledgement** does not undo the effect.

> Read the labels and values. Explain the relationship the model shows.

## 04. Change one thing. Trace what follows.



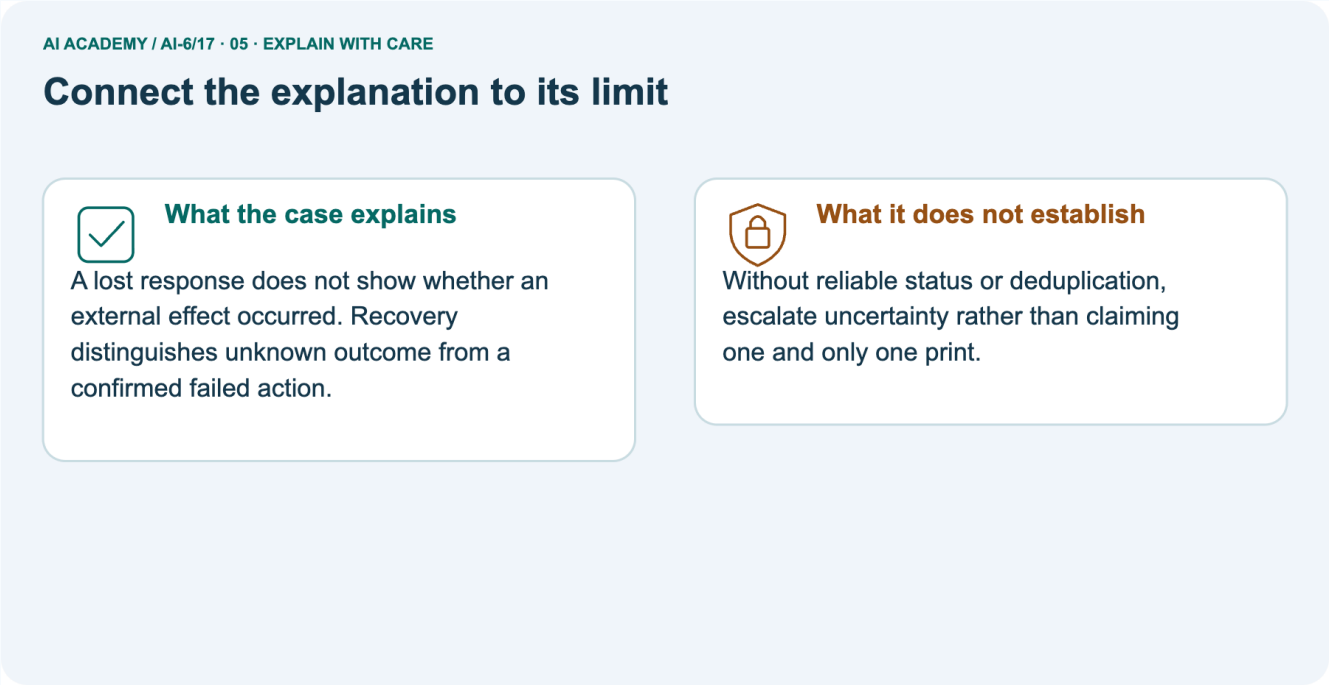

Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: If complete, return that result without issuing another print job. Changed condition: Status reliably reports J6 absent.

Expected result: A policy-approved resend using the same deduplication key may be considered. Reason: Confirmed absence supplies different **evidence** from silence.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

## 05. Connect the explanation to its limit



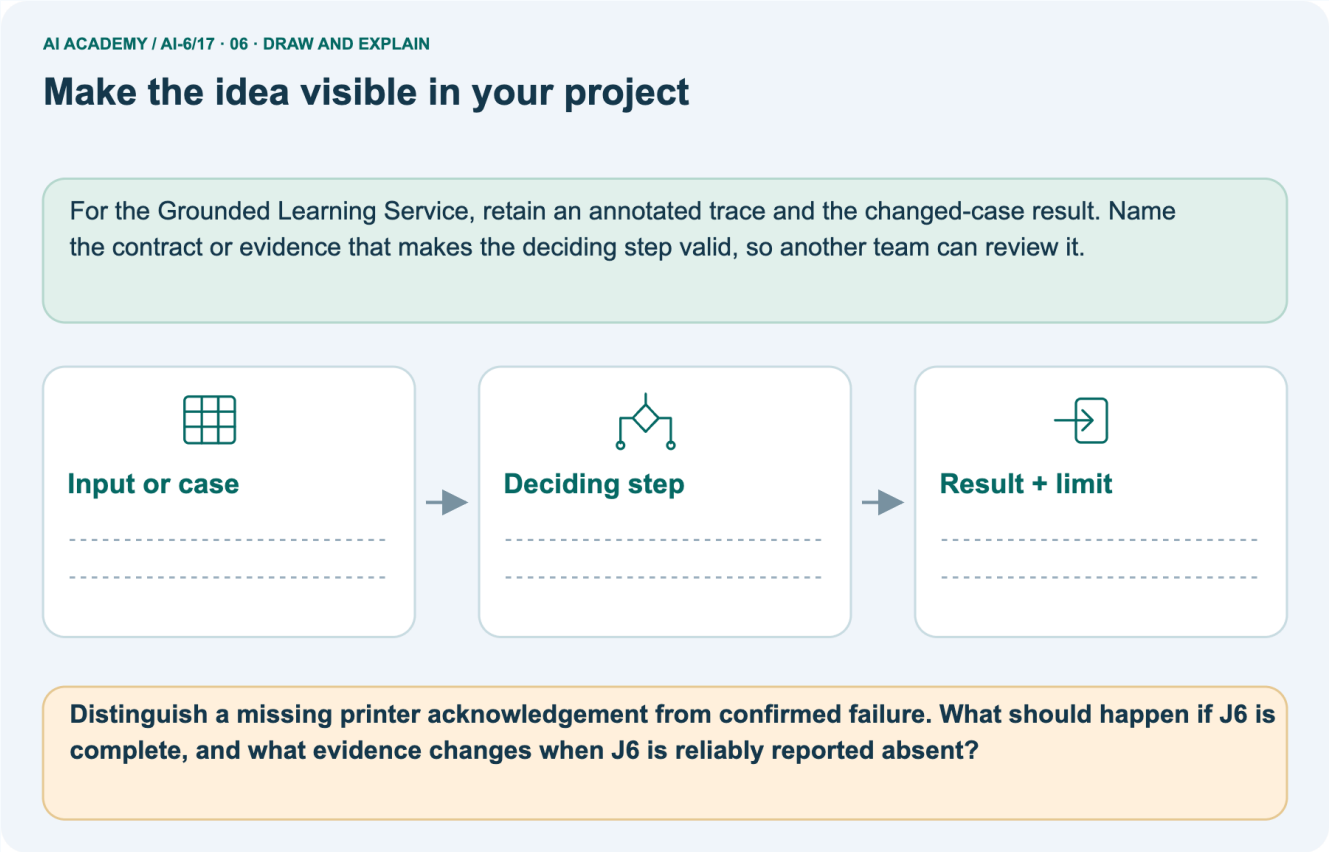

Follow the labelled picture with the matching explanation. Open the vector diagram

A lost response does not show whether an external effect occurred. Recovery distinguishes unknown outcome from a confirmed failed action.

Limit: Without reliable status or deduplication, escalate uncertainty rather than claiming one and only one print.

> Explain the case, then point to a limit on the claim.

## 06. Make the idea visible in your project




Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Distinguish a missing printer **acknowledgement** from confirmed failure.

What should happen if J6 is complete, and what **evidence** changes when J6 is reliably reported absent?

> Draw your project trace and justify the deciding step using **evidence**.

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.